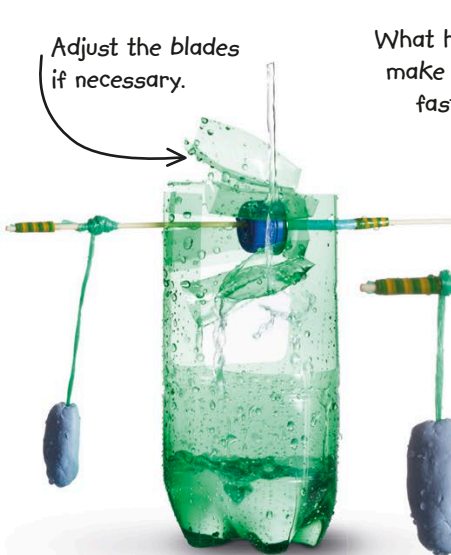




14 Tape a piece of string near the end of the skewer without the straw. Then press a lump of adhesive putty around the other end of the string—this will act as a weight.



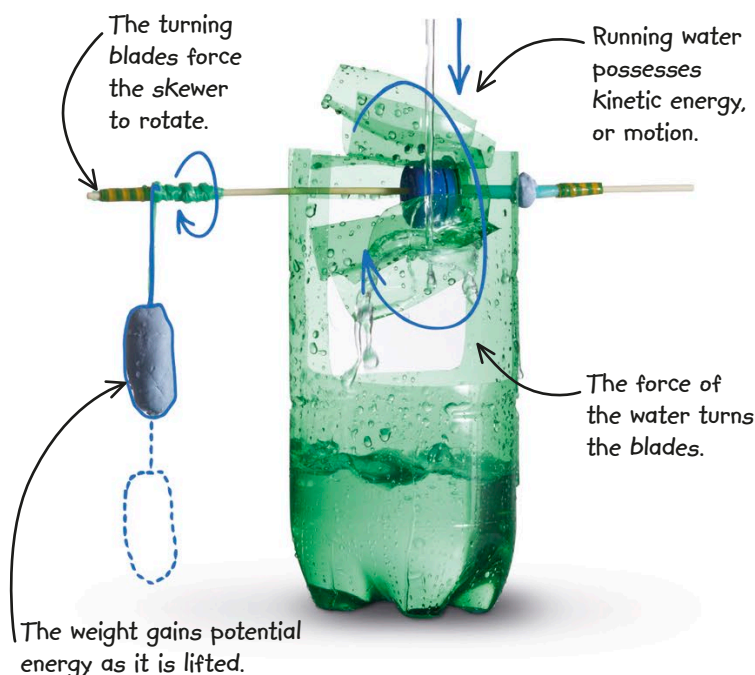
15 Now comes the fun part! Take your waterwheel outside or put it in a sink. Drip water on the wheel from a faucet or a measuring glass filled with water. The waterwheel should turn around and lift the putty.

What happens if you make the water run faster or slower?



HOW IT WORKS

When you pour the water, it exerts a force on the blades of the waterwheel, making them turn. The shaft exerts a force on the wooden skewer, turning it. As the skewer turns, it also applies a force on the attached string, which pulls up the adhesive putty weight.



REAL WORLD SCIENCE HYDROELECTRICITY



Flowing water can be used to generate electricity. In a hydroelectric power station, river water is held back by a dam, so that it builds up huge pressure and lots of potential energy. It flows under pressure through pipes, turning specially designed waterwheels called turbines. These, then, turn electric generators that supply many homes and businesses with electricity. This picture shows the top of the shafts of several turbines that spin horizontally. The generators are inside the round blue part at the top of each turbine.